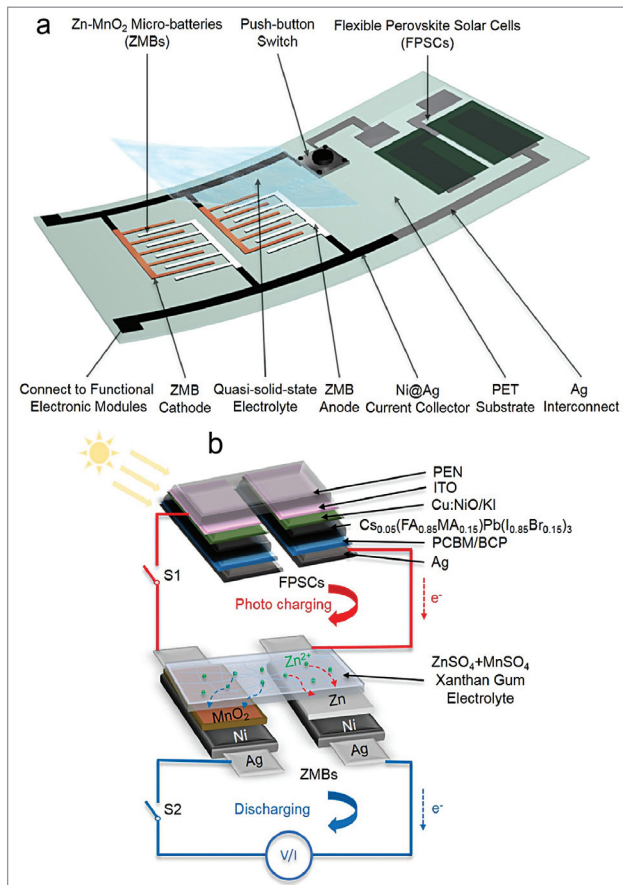


A battery that charges in sunlight within seconds

A renewable and rechargeable battery prototype created by the University of Surrey might extend the energy life of



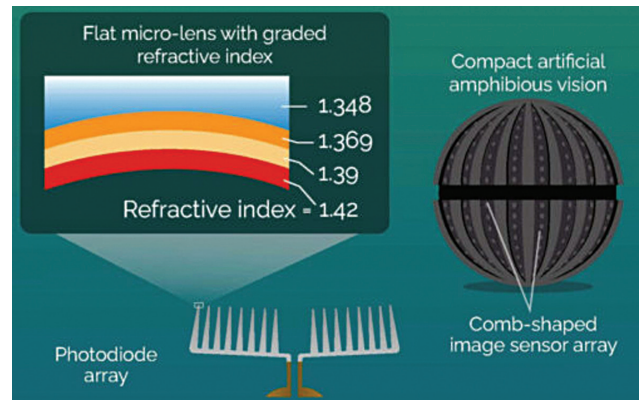
Schematics of the (a) device configuration and (b) working principle of the integrated flexible photo-rechargeable system (Credit: <https://techxplore.com>)

future smartwatches and other wearables by tens of minutes with just thirty seconds of sunshine. This was made possible by their innovative photo-rechargeable technology, which combines perovskite solar cells and zinc-ion batteries. The elegant and well-matched structural design between the integrated battery and solar cells makes this system stand out because it enables it to demonstrate high energy and volume density comparable to cutting-edge micro batteries and super capacitors. This can eventually promote many applications in self-powered wearable internet-of-things, autonomous power systems, and emergency electronics.

An artificial vision system inspired by a crab

There are significant drawbacks with present artificial vision systems, like hemispherical (180°) field-of-view and

inability to image both terrestrial and underwater surroundings. A team of researchers from the United States and Korea have now created a cutting-edge artificial vision system with omnidirectional imaging capabilities that can

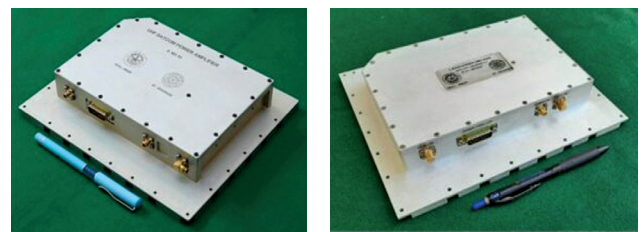


The novel vision system (Credit: <https://www.gist.ac.kr>)

operate in both aquatic and terrestrial environments. They literally recreated the Fiddler crab's eyes in an aquatic artificial vision system that was sharp and distortion-free. They demonstrated how the system had a panoramic sight field in both air and water that measured 300° horizontally and 160° vertically.

RF power amplifiers being developed indigenously

IIT Roorkee and the Defence Research and Development Organisation (DRDO) have partnered to research defence technology and produce homegrown RF power amplifiers. According to IIT Roorkee, these amplifier units perform better than average, attaining high efficiency and gain



Radio frequency power amplifier developed by DRDO and IIT Roorkee (Credit: ANI)

while also taking into account effective harmonic and intermodulation suppression. In fact, they were created to be able to meet the demands for high efficiency (for thermal management) and good linearity (for signal fidelity) at the same time. The amplifiers will be mass produced by private domestic partners and assembled with indigenous radio modules from Defence Electronics Application Laboratory (DEAL) and DRDO.